

Developing a Low-Cost IoT Prototype for Non-Invasive Elderly Monitoring[†]

Danilo Gomez-Giraldo ¹, Valerie Rios-Lopez ², Duver Arboleda-Marles ³, Esteban Gonzalez-Valencia ⁴, and Jorge Montoya-Cardona ^{4,*}

¹ Department of Mechatronics and Electromechanics, Engineering Faculty, Institución Universitaria ITM, Medellin, Colombia; danilogomez1129161@correo.itm.edu.co

² Department of Systems, Engineering Faculty, Institución Universitaria ITM, Medellin, Colombia; va-lerierios1155209@correo.itm.edu.co

³ Department of Administrative Sciences, Economic and Administrative Sciences Faculty, Institución Universitaria ITM, Medellin, Colombia; duverarboleda1142609@correo.itm.edu.co

⁴ Department of Electronics and Telecommunications, Engineering Faculty, Institución Universitaria ITM, Medellin, Colombia; estebangonzalez@itm.edu.co (E.G.-V.); jorgemontoyac@itm.edu.co (J.M.-C.).

* Correspondence: jorgemontoyac@itm.edu.co

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Abstract

Population aging has increased the need for affordable home-monitoring technologies that support older adults while preserving privacy. This work presents Radar-Cuidador, a low-cost, camera-free IoT prototype based on a distributed Hub-and-Spoke architecture integrating mmWave presence sensing, relative acoustic-event monitoring, environmental measurements, and bathroom-door activity. Sensor data are transmitted through Wi-Fi to ThingSpeak, where MATLAB-based rules evaluate predefined conditions and return an alert state for local actuation. Laboratory verification confirmed sensor acquisition, cloud transmission, control-channel writeback, and actuator response. Including two 3D-printed enclosures, the prototype cost approximately COP 142,000 (USD 46). The results establish functional feasibility but not emergency-detection accuracy or residential performance.

Keywords: Internet of Things (IoT); elderly monitoring; assisted living; mmWave radar; privacy-aware sensing.

1. Introduction

Population aging is creating new challenges for healthcare and social-support systems worldwide. The World Health Organization projects that the global population aged 60 years and older will increase from approximately 1 billion in 2020 to 1.4 billion in 2030 and 2.1 billion by 2050 [1]. This demographic transition has increased interest in technologies that can help older adults remain safely and independently in their own homes. Smart-home technologies may support aging in place through safety monitoring, emergency notification, chronic-condition management, and assistance with daily activities. However, their implementation continues to face challenges related to cost, technical reliability, privacy, digital literacy, and user acceptance [2,3].

In-home monitoring systems employ different sensing approaches, including wearable devices, cameras, motion detectors, contact sensors, environmental sensors, and other ambient technologies. A scoping review of in-home monitoring technologies identified passive infrared motion sensors and contact sensors among the most frequently used devices for

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observing presence, daily activities, and behavioral patterns [4]. Wearable devices can provide personal physiological or activity information, but their effectiveness depends on consistent use, charging, and adherence. These requirements may limit their usefulness for some community-dwelling older adults [5]. Ambient sensors offer a complementary approach because they can operate with little or no active interaction from the monitored person.

Camera-based systems can provide detailed contextual information for activity recognition and fall detection. Nevertheless, continuous visual monitoring in residential spaces may raise concerns regarding privacy, security, autonomy, data access, and user acceptance [6,7]. These concerns are especially relevant in private areas of a home and motivate the development of non-imaging sensing strategies that collect sufficient contextual information without recording identifiable visual data.

Millimeter-wave (mmWave) radar is a promising non-contact sensing technology because it can detect human presence and motion without conventional image capture and is not dependent on visible-light conditions. Advanced mmWave systems have demonstrated real-time human tracking, activity recognition, and fall-detection capabilities in controlled indoor environments [8,9]. Systematic evidence also shows that the reported performance of fall-detection technologies depends strongly on the sensor type, processing method, evaluation dataset, and experimental protocol [10]. Consequently, results obtained using advanced radar arrays and activity-classification pipelines cannot be transferred directly to low-complexity presence radars that provide only binary presence and energy-related outputs.

Acoustic sensing can contribute additional information about strong impacts or unusual events, but an isolated sound measurement does not establish that a fall has occurred. Previous in-home evaluation of a sound-based fall-detection system found that everyday sounds could generate frequent false-positive alerts, illustrating the difficulty of interpreting acoustic events without sufficient contextual information [11]. Combining acoustic measurements with presence, environmental, and contact-sensor information may therefore provide a more informative basis for preliminary rule-based alerts, although such combinations require controlled validation before their detection performance can be established.

The Internet of Things (IoT) provides a practical framework for integrating these complementary sensing modalities. Embedded devices with wireless connectivity can acquire measurements from different nodes, transmit them to cloud platforms, and support remote storage, visualization, and rule-based processing. A distributed architecture can also separate the sensing functions assigned to different areas while maintaining a shared monitoring and alert-management service.

This work presents Radar-Cuidador, a low-cost IoT prototype intended for non-invasive monitoring of older adults without cameras. The system follows a distributed Hub-and-Spoke architecture. An ESP32-based Main Hub integrates an HLK-LD2410C mmWave radar for presence and motion-related sensing, a MAX4466 microphone module for relative acoustic-event monitoring, and a BME280 sensor for ambient temperature and relative humidity. A secondary ESP8266-based Bathroom Spoke uses a magnetic reed switch to register door-state changes and the elapsed time since the most recent change. Both nodes transmit their data through Wi-Fi to ThingSpeak. A MATLAB script applies four predefined rules and writes the resulting alert state and event-type code to a control channel, which is periodically queried by the Main Hub to manage local acoustic and visual actuators.

The scope of this study is limited to the design, construction, integration, and laboratory-level functional verification of the prototype. It does not report a residential field

deployment, clinical evaluation, or tests of diagnostic performance in an older-adult population. The available evidence supports the acquisition and transmission of sensor variables, their visualization in ThingSpeak, and the implementation of a rule-based monitoring workflow. It does not establish detection accuracy, sensitivity, specificity, false-positive rate, or the ability to prevent or respond successfully to real domestic emergencies.

The main contribution of this work is the integration of multiple non-imaging sensors into a distributed IoT prototype that combines local acquisition, cloud communication, rule-based event assessment, control-channel feedback, and local alert actuation. By avoiding camera-based monitoring, the design adopts a privacy-aware sensing approach while preserving complementary information about presence, acoustic activity, environmental conditions, and door use. Radar-Cuidador should therefore be understood as an assisted-living proof of concept and not as a medical diagnostic device or a replacement for caregivers or professional supervision. The implemented architecture provides a foundation for future controlled experiments and subsequent evaluation in representative residential environments.

2. Radar-Cuidador System Design and Hardware Implementation

2.1. System Architecture

Radar-Cuidador was implemented using a distributed Hub-and-Spoke architecture comprising an ESP32-based Main Hub, an ESP8266-based Bathroom Spoke, and a cloud-mediated feedback path. The Main Hub acquires radar, acoustic, temperature, and relative-humidity variables and controls the local acoustic and visual alert actuators. The Bathroom Spoke independently records the state of a magnetic door sensor and the elapsed time since its most recent transition. The two embedded nodes do not exchange data directly; each connects through Wi-Fi and sends its measurements by HTTP to a dedicated ThingSpeak channel.

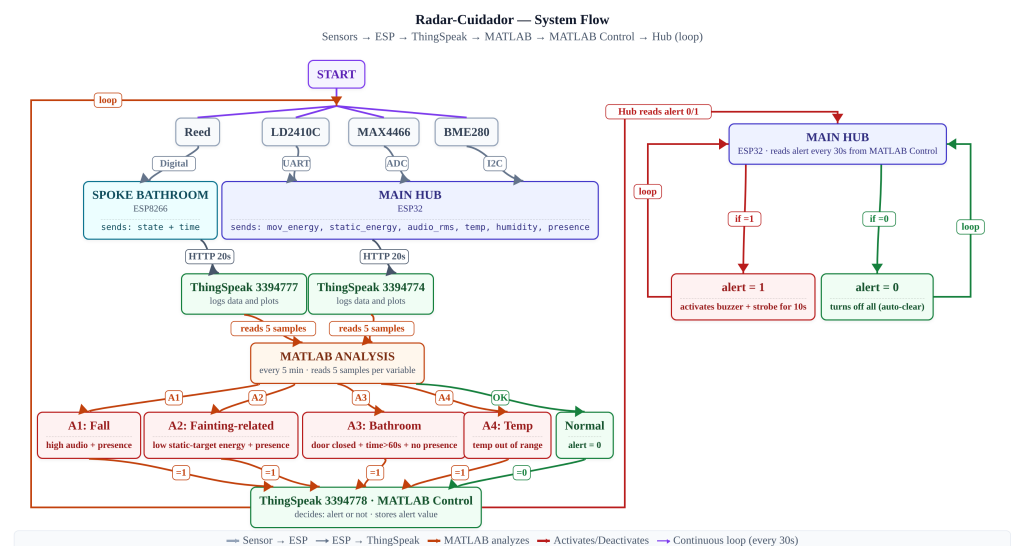


Figure 1. System architecture and end-to-end data flow of the Radar-Cuidador prototype, showing the Main Hub, Bathroom Spoke, ThingSpeak channels, rule-based evaluation, and alert-feedback path.

Figure 1 summarizes the end-to-end data path. ThingSpeak stores the records transmitted by both nodes, and a MATLAB Analysis script reads the relevant variables and applies the predefined rule set. The resulting binary alert state and event-type code are written to a separate control channel. The Main Hub periodically queries the alert field of this channel and updates the buzzer and stroboscopic indicator according to the received

state. This feedback path separates sensor acquisition from cloud-side rule evaluation while retaining local actuation at the Main Hub.

2.2. Hardware Components and Hub–Spoke Integration

The hardware implementation was distributed between the two embedded nodes according to their sensing and actuation functions. The Main Hub was built around an ESP32-WROOM-32, whose UART, analog-to-digital, I²C, and general-purpose input/output interfaces accommodate the sensors, status indicators, and alert actuators. The Bathroom Spoke uses an ESP8266 NodeMCU v3 to acquire the digital door-state signal and operate its local indicators. Table 1 summarizes the operating voltages, documented characteristics, and assigned functions of the components integrated into both nodes.

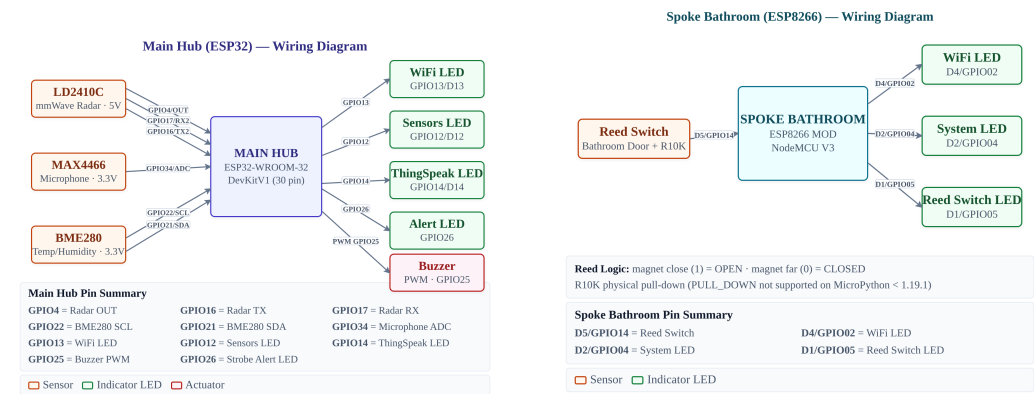
Table 1. Electrical and functional characteristics of the components used in the Radar-Cuidador prototype.

Component	Operating voltage	Documented specification or response	Measured variable or function
ESP32-WROOM-32 (Hub)	3.3 V	Clock frequency of up to 240 MHz, with digital, analog, UART, I ² C, and Wi-Fi interfaces	Processing of Hub sensors, Wi-Fi communication with ThingSpeak, and control of alert actuators
ESP8266 NodeMCU v3 (Spoke)	3.3 V	Microcontroller with integrated Wi-Fi, an 80 MHz clock frequency, and digital interfaces	Acquisition of the bathroom-door state and transmission of data to ThingSpeak
HLK-LD2410C (mmWave radar)	5 V	24 GHz radar with presence detection and moving- and static-target energy outputs ranging from 0 to 100	Binary presence (0/1), motion energy (0–100), and static-target energy (0–100)
BME280	3.3 V	Temperature accuracy: ± 1 °C; relative-humidity accuracy: ± 3 %RH; I ² C communication at address 0x76	Ambient temperature (°C) and relative humidity (%)
MAX4466	3.3 V	Amplified microphone with an analog response proportional to the incident sound level	Relative audio RMS level (0–4095) used together with radar presence in the rule-based fall-alert condition
Reed switch	— (switched)	Normally open magnetic switch with a binary response determined by the magnet position	Bathroom-door state (0 = closed, 1 = open)
Passive buzzer	5 V	Piezoelectric buzzer controlled by pulse-width modulation at 500 Hz	Local acoustic alert actuator
Wi-Fi indicator LED (2 units)	— (with 220 Ω resistor)	5 mm LED with a 220 Ω current-limiting resistor	Local indication of the Wi-Fi connection status of the ESP32 and ESP8266
Sensor indicator LED	— (with 220 Ω resistor)	5 mm LED with a 220 Ω current-limiting resistor and a heartbeat pattern every 10 s	Local indication of the operating status of the Main Hub sensors
ThingSpeak indicator LED (2 units)	— (with 220 Ω resistor)	5 mm LED with a 220 Ω current-limiting resistor, controlled after a ThingSpeak transmission attempt	Local indication of the cloud-communication status
Stroboscopic alert LED	— (with 220 Ω resistor)	5 mm LED with an ON/OFF pattern controlled every 300 ms	Visual alert actuator used in the laboratory prototype
220 Ω resistor and 10 k Ω resistor resistors	—	220 Ω resistors for the LEDs and a 10 k Ω physical pull-down resistor for the reed switch	LED current limitation and definition of the reed-switch logic level on the ESP8266

Note: The listed values correspond to nominal or documented component specifications used as design references. They do not represent calibration or metrological-validation results obtained in this study.

In the Main Hub, the HLK-LD2410C is connected through its digital OUT signal and the ESP32 UART2 interface. The MAX4466 provides an analog signal to the ESP32 analog-to-digital converter, whereas the BME280 communicates through I²C. Independent GPIO outputs control the Wi-Fi, sensor-status, ThingSpeak-status, and stroboscopic-alert LEDs, while the passive buzzer is driven by a pulse-width-modulated output.

The Bathroom Spoke connects the reed switch to a digital input of the ESP8266. A physical 10 k Ω pull-down resistor defines the input logic level, and three additional outputs control the Wi-Fi, system-status, and reed-switch indicators. The LEDs in both nodes incorporate 220 Ω current-limiting resistors. Figure 2 presents the interfaces and pin assignments used in the two circuits.



(a) (b)

Figure 2. Wiring and pin-assignment diagrams of the Radar-Cuidador nodes: (a) Main Hub based on the ESP32-WROOM-32; and (b) Bathroom Spoke based on the ESP8266 NodeMCU v3.

2.3. Prototype Assembly and 3D-Printed Enclosures

The Main Hub and Bathroom Spoke were assembled as separate breadboard-based modules, preserving the physical separation established by the Hub-and-Spoke architecture. After completing the component placement and wiring, each module was installed in an individual 3D-printed enclosure. The enclosures incorporate openings that keep the micro-USB connectors of the ESP32 and ESP8266 accessible and allow the externally visible indicators and components to remain exposed. Figure 3 documents the internal arrangement of both circuits and their final enclosed configuration.

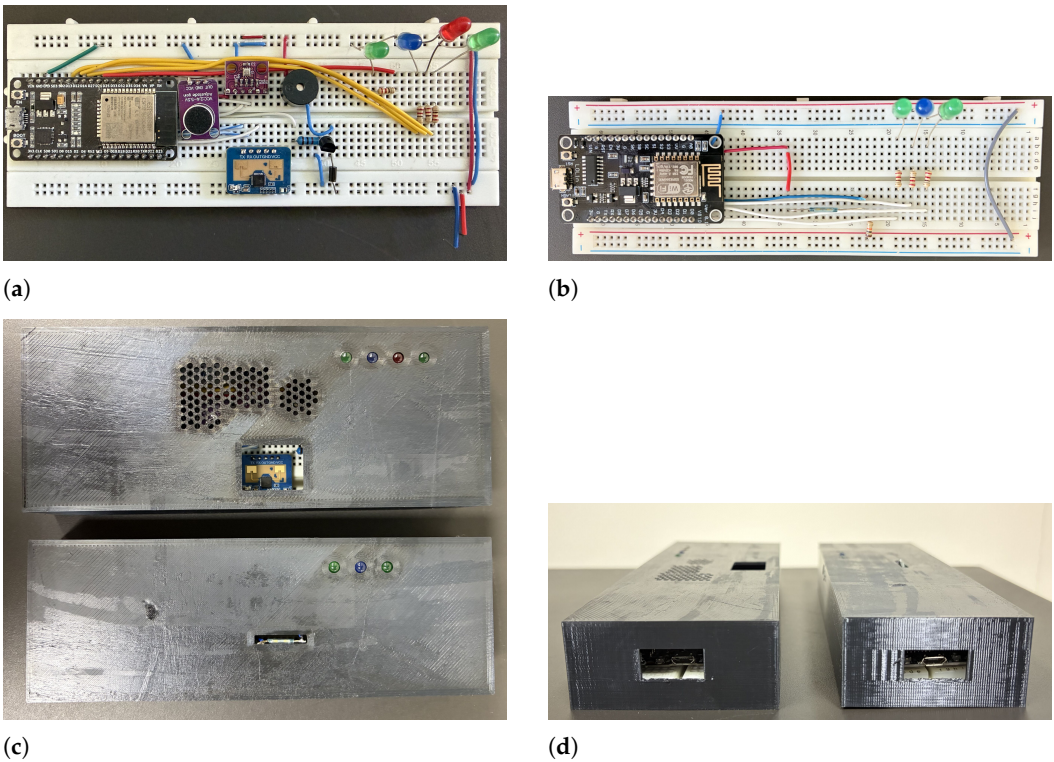


Figure 3. Radar-Cuidador prototype before and after enclosure: (a) unboxed Main Hub; (b) unboxed Bathroom Spoke; (c) top view of both nodes housed in their 3D-printed enclosures; and (d) front view showing the openings for the micro-USB connectors.

3. Firmware, Data Acquisition, and IoT Alert Management

3.1. Main Hub Sensor Acquisition and Local Actuation

The Main Hub firmware was implemented in MicroPython v1.28.0 on the ESP32. During initialization, the program configures the UART, analog-to-digital, I²C, pulse-width-modulation, and digital input/output interfaces required by the connected devices. The BME280 is initialized and checked through an initial reading before the program enters the continuous acquisition loop. Separate functions service the radar, microphone, and environmental sensor, allowing each input to be processed according to its communication interface and output format.

The HLK-LD2410C is read through its digital OUT signal and UART2 interface at 256,000 baud. The digital signal provides the binary presence state, while the UART routine searches the received buffer for the expected frame header and verifies that sufficient bytes are available before updating the motion- and static-target-energy variables. Incomplete or unrecognized frames are not used to update these values.

The MAX4466 output is acquired through the 12-bit analog-to-digital converter. For each acquisition, the firmware collects 100 samples separated by 200 μ s and updates an adaptive baseline using a smoothing factor of 0.02. The root-mean-square value of the sample deviations from this baseline is then calculated as a relative acoustic-activity indicator on the ADC scale. This value is not a calibrated sound-pressure measurement. To prevent the local buzzer from entering the acoustic measurement, the microphone reading is suppressed while the buzzer is active and for 2500 ms after its most recent activation.

Temperature and relative humidity are obtained from the BME280 through the I²C interface. A successful reading updates both environmental variables, whereas an acquisition exception activates the sensor-error flag used by the local status indicator. Local alert actuation is handled through a non-blocking routine that drives the passive buzzer using a 500 Hz pulse-width-modulated signal and alternates the stroboscopic LED at 300 ms intervals in response to the binary alert state. When no alert is active, both outputs remain off.

3.2. Bathroom Spoke and Door-State Monitoring

The Bathroom Spoke firmware was implemented in MicroPython v1.19.1 on the ESP8266. During initialization, the program configures the reed-switch input and the local indicator outputs, establishes the Wi-Fi connection, and obtains the initial door state before entering the main loop. The magnetic input uses the binary convention established for the prototype: 0 represents a closed door and 1 represents an open door.

During each loop iteration, the current reed-switch value is compared with the previously stored state. When a transition is detected, the firmware updates the door state, resets the elapsed-time counter, and stores the time of the change. While the state remains unchanged, the elapsed time is calculated from the most recent transition. The resulting variables represent the current door state and the number of seconds since its last change. The reed-switch indicator remains active when the door is closed, while the remaining local indicators report the connection and operating status of the node.

Because the Main Hub, Bathroom Spoke, local actuators, and cloud-side analysis operate at different intervals, Table 2 consolidates their configured execution conditions. The intervals are presented in the table rather than repeated in the descriptions of the individual routines.

Table 2. Timing parameters and execution conditions of the Radar-Cuidador acquisition, communication, and alert-management routines.

Routine or task	Configured interval	Execution condition
Main-loop iteration of the Hub	Approximately 1 s	Coordinates sensor acquisition, the sensor-status indicator, and the periodic control-channel query
LD2410C radar reading (UART)	Each loop iteration	Checks the OUT signal and, when a complete UART frame is available, updates the presence and energy variables
MAX4466 microphone reading (ADC)	100 samples per iteration with 200 μ s between samples	Calculates the RMS value and suppresses the microphone reading for 2500 ms after the most recent buzzer activation
BME280 reading (I ² C)	Each loop iteration	Updates the temperature and relative-humidity values and sets the sensor-error flag if the reading fails
Sensor LED heartbeat	10 s	Produces two short pulses when no sensor error is present and remains on when an error is registered
Control-channel reading	30 s	Requests the most recent value of <code>field1</code> and updates the binary alert state
Hub data transmission to ThingSpeak	20 s	Builds the configured Hub data payload and performs an HTTP POST when Wi-Fi is available
Main-loop iteration of the Spoke	Approximately 1 s	Coordinates the reed-switch reading, elapsed-time calculation, and local status indicators
Reed-switch reading	Each loop iteration	Compares the current and previous states and resets the elapsed-time counter when a transition occurs
System LED heartbeat (Spoke)	10 s	Produces three short pulses when no communication error is present and remains on when an error is registered
Spoke data transmission to ThingSpeak	20 s	Sends the door state and elapsed time through an HTTP POST and updates the communication-error flag if the request fails
Stroboscopic LED and buzzer update	300 ms toggle within a 10 s actuation window	Alternates the 500 Hz buzzer and the stroboscopic LED during the actuation window and disables both outputs after 10 s
MATLAB Analysis (TimeControl)	5 min	Reads the five most recent samples of the required variables, evaluates the four predefined rules, and writes the alert state and event-type code to the control channel

Note: The main-loop delay does not represent a calibrated sampling period because the effective execution time also depends on peripheral response and communication operations.

3.3. HTTP Communication and ThingSpeak Data Model

The Main Hub and Bathroom Spoke use the MicroPython `urequests` library to transmit their records to ThingSpeak through HTTP POST requests. Each request contains the write-authorization key of the destination channel and the variables produced by the corresponding node. The response object is closed after each completed request to release memory on the embedded device.

The ThingSpeak data model uses separate channels for the Main Hub measurements, the Bathroom Spoke records, and the control values generated by MATLAB Analysis. This organization keeps the measurements acquired by each node separate from the alert information returned to the Main Hub. Table 3 defines the correspondence between the source variables, HTTP parameters, and transmitted information. The processing of these records and the subsequent control-channel feedback are described in Section 3.4.

Table 3. Mapping between the Radar-Cuidador variables and the fields of the ThingSpeak data channels.

HTTP parameter	Channel	Source variable	Transmitted information
api_key	3394774 (Hub)	TS_HUB_WRITE	Write-authorization key of the Main Hub channel
field2	3394774 (Hub)	mov_energy	Motion energy reported by the radar (0–100)
field3	3394774 (Hub)	static_energy	Static-target energy reported by the radar (0–100)
field4	3394774 (Hub)	audio_rms	Relative audio RMS level obtained from the MAX4466 (0–4095)
field5	3394774 (Hub)	temperature	Ambient temperature obtained from the BME280 (°C)
field6	3394774 (Hub)	humidity	Relative humidity obtained from the BME280 (%)
field7	3394774 (Hub)	presence	Binary presence state reported by the radar (0/1)
field8	3394774 (Hub)	active_alert	Alert state retrieved by the Main Hub from the control channel (0/1)
api_key	3394777 (Bathroom)	TS_WRITE	Write-authorization key of the Bathroom Spoke channel
field1	3394777 (Bathroom)	door_state	Bathroom-door state (0 = closed, 1 = open)
field2	3394777 (Bathroom)	seconds_since_change	Time elapsed since the most recent door-state transition (s)
api_key	3394778 (Control)	controlWriteKey	Write-authorization key used by MATLAB Analysis for the control channel
field1	3394778 (Control)	alert	Binary alert state written by MATLAB Analysis (0 = normal, 1 = alert)
field2	3394778 (Control)	emergency_type	Event-type code generated by the rule-based analysis (0–4)

Note: field1 of the Main Hub channel is not used in the implemented data model. The channel credentials are not reported in the article. If a network request cannot be completed, local acquisition continues, but the corresponding record is not queued for later retransmission.

3.4. Rule-Based Alert Logic and Feedback Loop

The final stage of the monitoring workflow is implemented through MATLAB Analysis on ThingSpeak. The analysis is scheduled through TimeControl and reads the five most recent samples of the required Main Hub and Bathroom Spoke variables every 5 min. Before applying a condition, the script verifies that the corresponding data arrays are not empty. The binary alert state and event-type code are initialized to zero at the beginning of each execution.

Four threshold-based conditions are then evaluated. Condition A1 assigns event type 1 when the maximum relative audio RMS level exceeds 3000 and the mean radar-presence value is greater than 0.5. Condition A2 assigns event type 2 when the mean static-target energy is below 10 while the mean presence value remains greater than 0.5. Condition A3 assigns event type 3 when the bathroom door is predominantly closed, the elapsed time since its most recent state change exceeds 60 s, and the mean presence value in the Main Hub area is below 0.5. Condition A4 assigns event type 4 when the maximum temperature exceeds 38 °C or the minimum temperature falls below 10 °C. These thresholds correspond to the configuration of the laboratory prototype and are not clinically validated decision limits.

The four conditions are evaluated in a fixed priority order using an if-elseif structure. The first satisfied condition sets the binary alert value to 1 and assigns its corresponding event-type code, after which the remaining conditions are not evaluated during that execution. If none of the conditions is satisfied, both values remain at zero. At the end of each execution, MATLAB Analysis writes the alert state and event-type code to field1 and field2 of the MATLAB Control channel, respectively, following the mapping presented in Table ??.

The Main Hub queries the most recent binary alert value from the control channel every 30 s. When an active alert is received, the firmware starts a 10 s actuation window in which the passive buzzer operates at 500 Hz and the stroboscopic LED alternates every 300 ms. Both outputs are disabled after this interval, even if the stored control value remains equal to 1. When the control channel returns to zero, the actuation timer is reset so that a subsequent alert can initiate a new 10 s cycle. The event-type code remains stored in

ThingSpeak but is not required by the Main Hub to activate the local outputs. The resulting acquisition, evaluation, feedback, and actuation sequence is summarized in Figure 1.

4. Results and Discussion

4.1. Laboratory Setup and Functional Verification

The completed prototype was subjected to component-level and integrated functional checks under laboratory conditions. For the Main Hub, movement in front of the LD2410C produced changes in the presence and energy outputs, while controlled claps and impacts near the MAX4466 produced variations in the relative audio RMS value. The BME280 returned ambient temperature and relative-humidity measurements throughout the tests. For the Bathroom Spoke, moving the magnet toward and away from the reed switch produced the two configured door states and reset the elapsed-time counter after each transition.

The local alert outputs were also checked with the binary alert state active. The passive buzzer and stroboscopic LED followed the configured actuation pattern and returned to their inactive state at the end of the actuation window. These checks verified the response of the individual sensing paths and local actuators under the laboratory conditions used for the prototype.

The documented hardware budget was COP 122,000. Adding two 3D-printed enclosures at COP 10,000 each resulted in an estimated implementation cost of COP 142,000, equivalent to approximately USD 46 at the exchange rate available during manuscript preparation. Table 4 presents the corresponding cost breakdown.

Table 4. Engineering budget breakdown for the functional Radar-Cuidador prototype.

Component	Quantity	Unit cost (COP)	Subtotal (COP)
ESP32 microcontroller (Main Hub)	1	25,000	25,000
ESP8266 microcontroller (Bathroom Spoke)	1	10,000	10,000
LD2410 mmWave radar	1	35,000	35,000
MAX4466 microphone	1	8,000	8,000
BME280 environmental sensor	1	15,000	15,000
Reed switch and magnet	1	3,000	3,000
Alert actuators (passive buzzer and stroboscopic LED)	1	10,000	10,000
Breadboards, cables, and power supply	1	16,000	16,000
3D-printed enclosures	2	10,000	20,000
Total estimated cost			142,000

4.2. End-to-End Data Transmission and Visualization Results

ThingSpeak received and retained the laboratory records generated by the Main Hub, as shown in Figure 4. The radar measurements in Figure 4(a) present variations in motion and static-target energy together with the binary presence state. The energy signals exhibit different combinations during intervals with and without presence, showing that these variables were transmitted as independent radar outputs rather than being reduced to the binary presence indicator.

Figure 4(b) compares the relative audio RMS level with the presence state. The audio measurements remained below the configured threshold of 3000 throughout the displayed sequence, including the intervals in which presence was active. Consequently, the plotted records do not contain the high-audio condition required for a fall-related alert. The environmental records in Figure 4(c) show gradual variations in temperature and relative humidity, interrupted by brief discontinuities after which both signals resume their previous progression. The visualization therefore preserves both the observed environmental variations and the discontinuities present in the laboratory records.

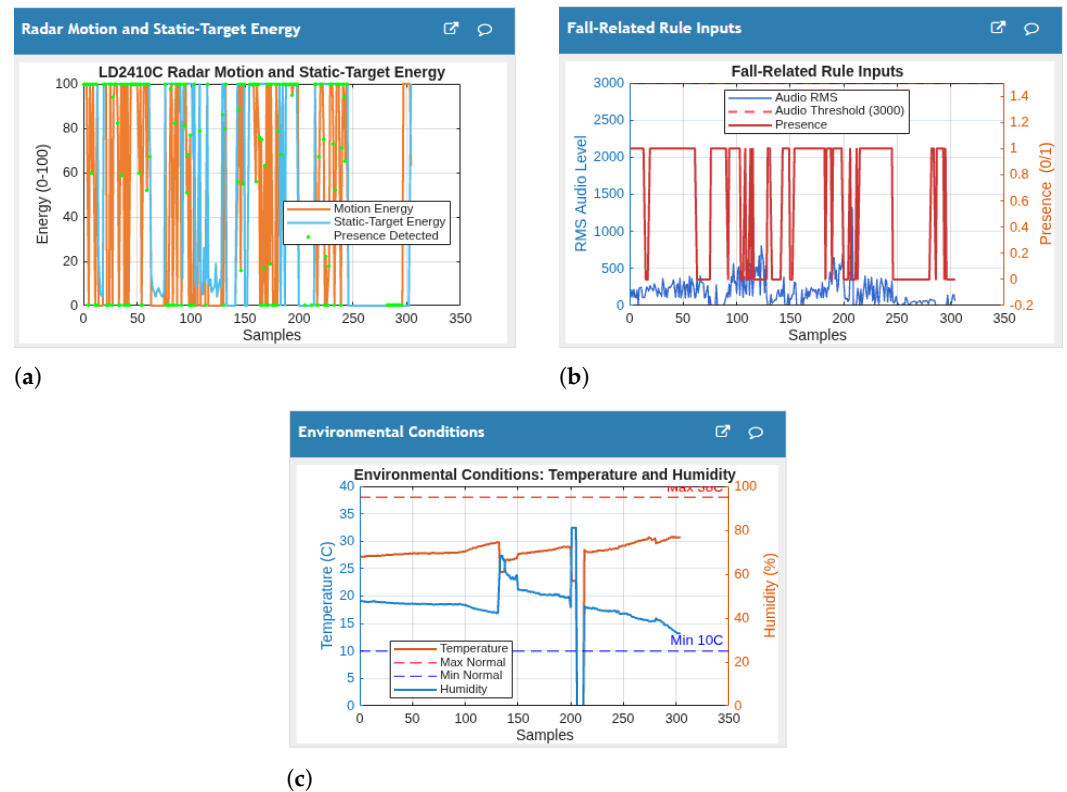


Figure 4. ThingSpeak visualizations of Main Hub laboratory records: (a) radar motion energy, static-target energy, and binary presence; (b) relative audio RMS level, configured threshold, and binary presence; and (c) ambient temperature and relative humidity.

The Bathroom Spoke records are presented in Figure 5. The binary value in Figure 5(a) remains equal to 1 during the displayed interval, corresponding to the documented open-door state. Figure 5(b) shows the elapsed-time field increasing between successive resets and reaching approximately 700 s during its longest displayed sequence. These records demonstrate that ThingSpeak stored both the direct reed-switch state and the elapsed-time variable calculated by the Bathroom Spoke.

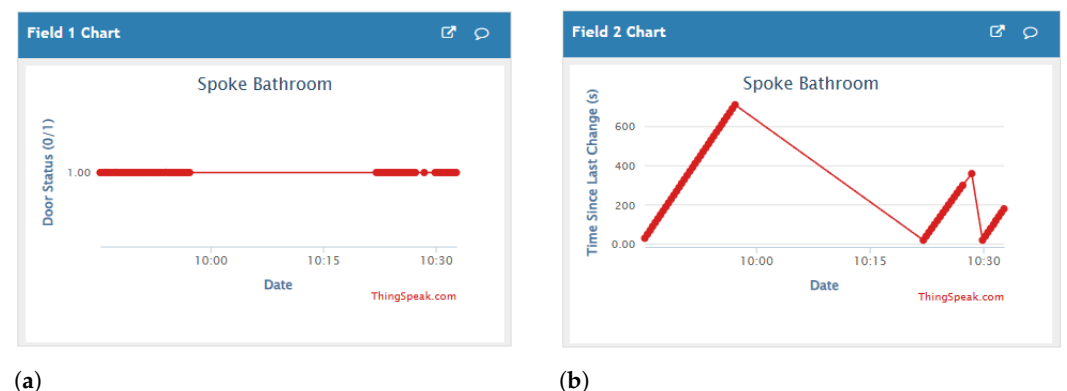


Figure 5. ThingSpeak visualizations of Bathroom Spoke laboratory records: (a) binary door state; and (b) elapsed time since the most recent state change.

Figure 6 presents the two variables stored in the MATLAB Control channel. Repeated zero-valued records appear in both the binary alert field shown in Figure 6(a) and the event-type field shown in Figure 6(b). These values indicate that the rule evaluations represented in the displayed interval ended with no active alert and an event-type code of

zero. Thus, Figure 6 documents normal-state writeback to the control channel rather than an active-alert episode.

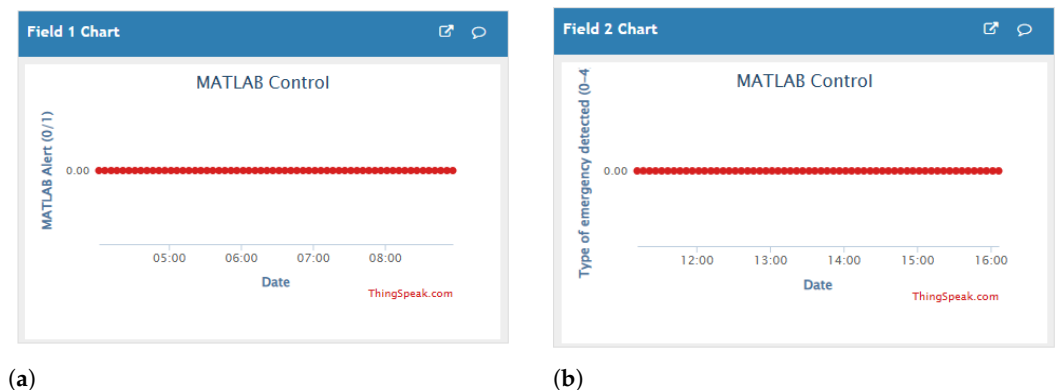


Figure 6. Records stored in the MATLAB Control channel during laboratory operation: (a) binary alert status; and (b) event-type code.

Taken together, Figures 4–6 document the arrival and visualization of the Main Hub and Bathroom Spoke measurements and the storage of the two resulting control variables in their corresponding ThingSpeak channels.

4.3. Limitations and Improvement Priorities

The experimental evidence is limited to functional verification under laboratory conditions. No tests were conducted in occupied residential environments or with older adults, and the evaluation did not use a labeled dataset, a reference monitoring system, or a repeated-event protocol. Consequently, the available records cannot be used to calculate sensitivity, specificity, false-alarm rate, detection latency, or long-term reliability. A controlled validation protocol should therefore define reproducible event scenarios, synchronized reference labels, repeated trials, and acceptance criteria before evaluating the prototype in a representative assisted-living environment.

The implemented configuration covers only the area assigned to the Main Hub and the Bathroom Spoke. The initial design considered additional monitored areas, but only one ESP8266-based Spoke was incorporated into the functional prototype. Extending the system will require additional nodes and an evaluation of wireless coverage, power distribution, and channel management as the network grows.

The breadboard-based construction also affected hardware stability during development. Loose connections, shared-ground problems, limited mounting space, and voltage drops produced intermittent or erratic sensor readings. One ESP32 board also developed a Wi-Fi hardware failure and had to be replaced. A custom printed circuit board, secure connectors, and a regulated power stage with a dedicated 5 V supply for the mmWave radar are the main priorities for improving electrical and mechanical reliability.

Cloud dependence introduces additional operational constraints. Measurements that fail during an HTTP transmission are not stored for subsequent delivery, and the rule evaluation depends on Wi-Fi connectivity and the availability of ThingSpeak. The 5 min MATLAB Analysis schedule also introduces a delay between sensor acquisition and cloud-side evaluation. Local buffering with timestamps, acknowledgment-based retransmission, and an edge-level fallback for time-sensitive alerts should be considered in a subsequent implementation.

The four alert conditions use thresholds selected for prototype testing rather than clinically validated decision limits. The current implementation assigns a single event-type code according to a fixed priority order, so simultaneous conditions are represented only

by the first matching rule. Further development should validate both the thresholds and the adopted priority policy using labeled experimental records or support simultaneous event codes when multiple conditions occur.

Finally, the current communication configuration was implemented for laboratory development. Before any deployment involving residential activity data, the system should define an appropriate channel-access policy and employ encrypted communication to protect the transmitted information.

5. Conclusions

This work established Radar-Cuidador as a functional, low-cost, camera-free IoT prototype for privacy-aware monitoring in assisted-living research. Its main contribution is the integration of mmWave presence and motion-related sensing, relative acoustic-event monitoring, ambient temperature and humidity measurements, and bathroom-door activity within a distributed Hub-and-Spoke architecture. The implemented workflow combines embedded acquisition, separate ThingSpeak data channels, cloud-side rule evaluation, control-channel feedback, and local acoustic and visual actuation. The completed two-node prototype, including its 3D-printed enclosures, had an estimated implementation cost of COP 142,000, equivalent to approximately USD 46 during manuscript preparation.

Laboratory verification showed that the Main Hub and Bathroom Spoke acquired their assigned variables and transmitted them to their corresponding ThingSpeak channels for storage and visualization. The available records also confirmed normal-state writeback to the MATLAB Control channel, while separate functional checks established that the passive buzzer and stroboscopic LED responded to an active binary alert. These findings demonstrate the operation of the implemented sensing, communication, processing, and actuation functions as a proof of concept; they do not establish emergency-detection accuracy or effectiveness in occupied residential environments. The next development stage should consolidate the electronics into a more robust hardware platform and conduct a controlled, labeled, and repeatable validation campaign before evaluating the system with older adults or in representative assisted-living environments.

Supplementary Materials: The following supporting information is publicly available at <https://github.com/Tgogi7/Developing-a-Low-Cost-IoT-Prototype-for-Non-Invasive-Elderly-Monitoring>: Software S1, ESP32 Main Hub firmware and supporting BME280 driver; Software S2, ESP8266 Bathroom Spoke firmware for door-state acquisition, elapsed-time calculation, and ThingSpeak communication; Software S3, MATLAB Analysis script implementing the four rule-based alert conditions and the alert-state and event-type writeback to the MATLAB Control channel.

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Data Availability Statement: The laboratory sensor records supporting the results presented in this study are publicly available through the ThingSpeak channels used by the prototype: Main Hub (<https://thingspeak.mathworks.com/channels/3394774>), Bathroom Spoke (<https://thingspeak.mathworks.com/channels/3394777>), and MATLAB Control (<https://thingspeak.mathworks.com/channels/>

3394778). The source code is available in the GitHub repository identified in the Supplementary Materials. No personal or clinical data were collected in this study.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

3D	Three-dimensional;
ADC	Analog-to-digital converter;
COP	Colombian peso;
GPIO	General-purpose input/output;
HTTP	Hypertext Transfer Protocol;
I ² C	Inter-Integrated Circuit;
IoT	Internet of Things;
LED	Light-emitting diode;
mmWave	Millimeter wave;
RH	Relative humidity;
RMS	Root mean square;
UART	Universal Asynchronous Receiver-Transmitter;
USB	Universal Serial Bus;
USD	United States dollar.

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